8(a)(i) Resistance $R = \frac{\rho l}{A} = \frac{(1.2\times10^{-5})(0.72)}{\pi(6.0\times10^{-2}\times10^{-3})^2}$

$= 764\ \Omega$

(ii) Power $P = \frac{V^2}{R} = \frac{230^2}{764}$ [1]

$= 69.2$ W

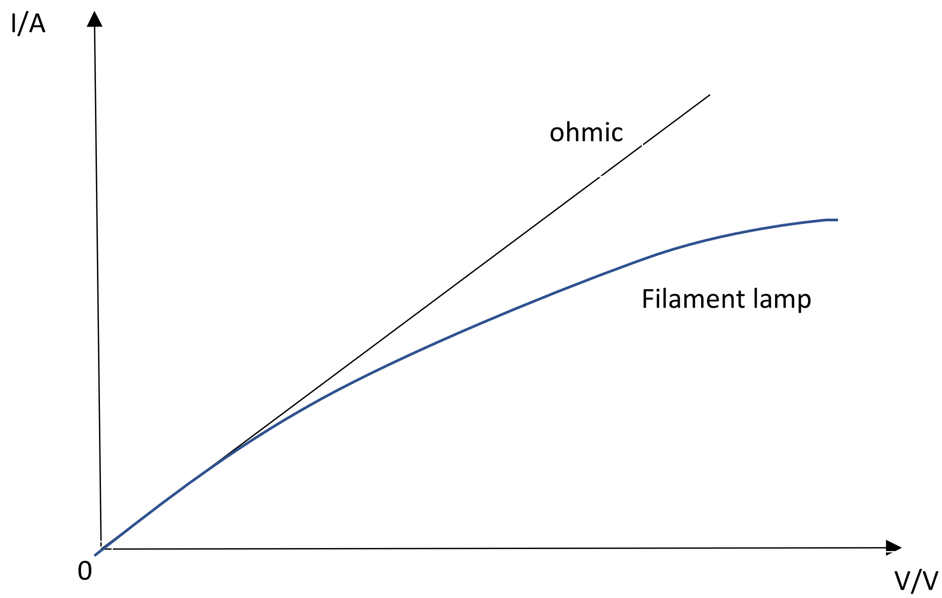
(iii) Material from filament will vaporise at high temperature.

(iv) 1. As filament gets thinner, its cross sectional area becomes effectively smaller [APPLICATION].

Since resistance α 1/cross sectional area [CONCEPT], resistance increases [CONCLUSION].

2. Power output is inversely proportional to resistance ($P = \frac{V^2}{R}$) [CONCEPT]. Since resistance of bulb increases [APPLICATION], power will drop as voltage V is constant [CONCLUSION].

(v)



(b) (i) 1. 2.10 V

2. $W = QV = (1)\,(2.10) = 2.10$ J

3. No current flows when switch is open and voltmeter is ideal.

No energy is being generated.

(ii) p.d. across internal resistor = E – V = 2.10 -2.00 = 0.10 V

By potential divider principle [CONCEPT],

Voltage across internal resistor = $\frac{r}{r+R}E$

$$0.10 = \frac{r}{r+10}(2.10)$$

$$r = 0.50\ \Omega$$

(iii) Efficiency = $\frac{I^2R}{I^2(R+r)} = \frac{R}{R+r} = \frac{1}{1+\frac{r}{R}}$ [CONCEPT]

As R increases, $1 + \frac{r}{R}$ decreases [APPLCATION], hence efficiency increases [CONCLUSION].